The coordinates of the image of the point (x_1, y_1) with respect to the line $ax + by + c = 0$ can be obtained by the line

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = -\frac{2(ax_1 + by_1 + c)}{a^2 + b^2}$$

Pair of straight lines	Condition for parallel	Condition for perpendicular
$ax^2 + 2hxy + by^2 = 0$	$h^2 - ab = 0,$	$a + b = 0$

The equation of the bisectors of the angle between the lines $ax^2 + 2hxy + by^2 = 0$ is

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

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The condition that the general second degree equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ should represent a pair of straight lines is $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

(i) The angle between them is $\theta = \tan^{-1}\left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$

If $a + b = 0$, then the lines are perpendicular.

(ii) The point of intersection

$$P\left(\frac{hf - bg}{ab - h^2}, \quad \frac{gh - af}{ab - h^2}\right)$$

(iii) Two straight lines represented by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ are parallel if $\frac{a}{h} = \frac{h}{b} = \frac{g}{f}$ or $bg^2 = af^2$

(iv) If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of parallel straight lines, then the distance between them is $2\sqrt{\frac{(g^2 - ac)}{a(a + b)}}$ or $2\sqrt{\frac{(f^2 - bc)}{b(a + b)}}$